Sparse Deletion and Triangular Prefix Sums

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Abstract

Let $E \subseteq \mathbb{N}$ have natural density zero, and let $a_0 < a_1 < a_2 < \cdots$ enumerate the positive integers outside E. We prove that the set of indices n for which

$$\sum_{i=0}^{n-1} a_i^2$$

is a triangular number has natural density zero. The result permits an arbitrary density-zero deletion set; no regularity hypothesis is imposed on its elements or their locations. The proof combines a finite-field sieve, the Chinese remainder theorem, a sparse-perturbation principle, and a transfer from value indices to indices in the retained enumeration. A Lean 4/Mathlib formalization of the theorem, including an axiom audit with no use of `sorryAx`, is available at `https://github.com/pczerkas/sparse-deletion-lean`.

Keywords. natural density; triangular numbers; sums of squares; finite fields; Chinese remainder theorem; formal verification; Lean.
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1 Introduction

A nonnegative integer s is *triangular* if

$$2s = m(m+1)$$

for some $m \in \mathbb{N}$. Write $\mathrm{Tri}(s)$ for this condition. For a set $A \subseteq \mathbb{N}$, let

$$A(x) = |A \cap \{0, 1, \ldots, x-1\}|.$$

We say that A has natural density zero if $A(x)/x \to 0$ as $x \to \infty$.

The purpose of this note is to establish a stability result for triangular prefix sums of squares under arbitrary sparse deletion. The point is that a density-zero deletion can have very large gaps and can delete very large integers; the argument does not require a quantitative bound on either phenomenon.

Theorem 1 (Sparse deletion theorem)**.** *Let $E \subseteq \mathbb{N}$ have natural density zero, and enumerate the positive integers outside E increasingly as*

$$a_0 < a_1 < a_2 < \cdots .$$

Then

$$\mathrm{d}\left\{ n \in \mathbb{N} : \mathrm{Tri}\left(\sum_{i=0}^{n-1} a_i^2 \right) \right\} = 0.$$

For $n = 0$, the sum is interpreted as 0, which is triangular. This isolated index clearly does not affect the conclusion.

Remark 2. Theorem 1 is a density statement, not a finiteness statement: it neither asserts that only finitely many triangular prefix sums occur nor that they eventually disappear. It also does not apply directly to prefix sums over primes, since the complement of the primes is not a density-zero deletion set.

2 Proof architecture

We record the ingredients of the proof, both to explain the theorem and to make clear the division between its arithmetic and density-theoretic parts. The complete formal proof is in the accompanying Lean repository.

2.1 A cubic modular obstruction

Put

$$S(x) = \sum_{k=0}^{x} k^2 = \frac{x(x+1)(2x+1)}{6}.$$

After an affine change of variables over a finite field of characteristic neither 2 nor 3, the condition that $S(x)+b$ be triangular is controlled by whether a value of a cubic polynomial is a square. The relevant cubic is affinely equivalent to

$$t \longmapsto t^3 - t.$$

The required value-set estimate is elementary.

Lemma 3 (Cubic value-set bound)**.** *Let $\mathbb{F}_q$ be a finite field of characteristic different from 3. Then*

$$|\{t^3 - t : t \in \mathbb{F}_q\}| \geq \frac{2q-1}{3}.$$

In the formal development, Lemma 3 follows by studying collisions of $t^3 - t$. Distinct elements in a common fibre give points on the conic

$$x^2 + xy + y^2 = 1,$$

and an elementary injection bounds the number of such collisions. No Weil bound, Hasse bound, elliptic-curve estimate, or character-sum estimate is used.

Together with the elementary count of quadratic residues in $\mathbb{F}_p$, this implies the following uniform local sieve bound. There is an infinite sequence of pairwise distinct primes $p \geq 37$ such that, for every integer shift b, at most a proportion $6/7$ of residue classes modulo p can satisfy the necessary modular condition for $S(x) + b$ to be triangular. The same estimate holds after translating x by any integer.

2.2 CRT amplification

Choose r distinct primes from the preceding family. Since their moduli are pairwise coprime, the Chinese remainder theorem multiplies the local bounds. Thus, for every integer shift b, there is a positive block length M such that every interval $[u, u + M)$ contains at most

$$\left(\frac{6}{7}\right)^r M$$

integers x for which $S(x) + b$ can be triangular. Crucially, M works uniformly for the starting point u and for the shift b.

2.3 Sparse perturbations

For the deletion set E, define

$$D(x) = -\sum_{\substack{0 \le k \le x \\ k \in E}} k^2.$$

Then the sum of squares of retained positive integers up to x is represented by $S(x)+D(x)$, up to the harmless indexing convention at 0. The function D is locally constant away from E: its increment can occur only at a deleted integer. The following abstract statement supplies the density step.

Lemma 4 (Sparse-perturbation principle)**.** *Let $E \subseteq \mathbb{N}$ have natural density zero. Let $f, d : \mathbb{N} \to \mathbb{Z}$ and $A \subseteq \mathbb{Z}$, and suppose that $d(n+1) = d(n)$ whenever $n+1 \notin E$. Assume that, for every $\varepsilon > 0$, there are $M > 0$ and $C \in \mathbb{N}$ with $C/M < \varepsilon$ such that, for all $u \in \mathbb{N}$ and $b \in \mathbb{Z}$,*

$$|\{n \in [u, u+M) : f(n) + b \in A\}| \le C.$$

Then

$$\mathrm{d}\{n \in \mathbb{N} : f(n) + d(n) \in A\} = 0.$$

Apply Lemma 4 with $f = S$, with the perturbation D, and with A the set of triangular integers. The CRT estimate makes C/M arbitrarily small by increasing r, hence triangular hits have density zero when indexed by their original values.

Finally, a density-zero deletion leaves sufficiently many integers: the n-th retained positive integer is eventually less than $2n+2$. An injection from triangular hits in the retained enumeration to the corresponding hits at value indices transfers density zero to the index set in Theorem 1. This completes the proof.

3 A concrete corollary

Let $c_0 < c_1 < c_2 < \cdots$ enumerate the composite positive integers. Since the set consisting of 1 and all primes has natural density zero, Theorem 1 yields the following consequence.

Corollary 5. *The set*

$$\left\{ n \in \mathbb{N} : \mathrm{Tri}\left(\sum_{i=0}^{n-1} c_i^2 \right) \right\}$$

has natural density zero.

4 Formal verification and reproducibility

The formalization is written in Lean 4 [1] and uses Mathlib [2]. The repository pins Lean and Mathlib to version `v4.35.0`; the exact dependency revisions are recorded in `lake-manifest.json`. The main formal theorem is

```
SparseDeletion.main_sparse_deletion_unconditional
    (E : Set Nat) (hE : HasDensityZero E) :
    HasDensityZero {n : Nat | Triangular (keptIndexPrefix E n)}
```

The repository provides separate verification targets for the end-to-end theorem and the generic sparse-perturbation lemma. It also contains targets for the exact finite-field value-set formula and the composite-number corollary. With Lean/Lake installed, run:

```
git clone https://github.com/pczerkas/sparse-deletion-lean.git
cd sparse-deletion-lean
lake exe cache get
lake build
```

The source includes axiom-printing certificate files. The principal proof terms use only the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`; they do not depend on `sorryAx`. As usual, formal verification establishes correctness relative to the Lean kernel, the pinned dependency code, and the accuracy of the formalized statement.

The repository also formalizes an optional, stronger classical finite-field identity: for $q \equiv 1 \pmod 3$ (respectively $q \equiv 2 \pmod 3$), the image of $t \mapsto t^3 - t$ in $\mathbb{F}_q$ has cardinality $(2q+1)/3$ (respectively $(2q-1)/3$). This exact formula is not needed for Theorem 1.

The source bundle includes `figure-proof-architecture.txt`, a plain-text dependency sketch suitable for conversion into an editorial figure if desired.

References

References

[1] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, pages 625–635, Springer, 2021. `https://doi.org/10.1007/978-3-030-79876-5_37`.

[2] The mathlib Community. The Lean Mathematical Library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381, ACM, 2020. `https://doi.org/10.1145/3372885.3373824`.

Availability and licence

The complete Lean source, build instructions, and dependency audit are publicly available at

`https://github.com/pczerkas/sparse-deletion-lean`.

This manuscript is licensed under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence. The accompanying Lean software repository is distributed separately under its repository licence (MIT at the time of this manuscript).

AI-use declaration

AI systems, including OpenAI ChatGPT, were used for mathematical exploration, proof development, refactoring, and Lean assistance. The author reviewed the mathematical statement and the formal development. The machine-checkable formal proof and its pinned dependencies are available in the repository cited above.